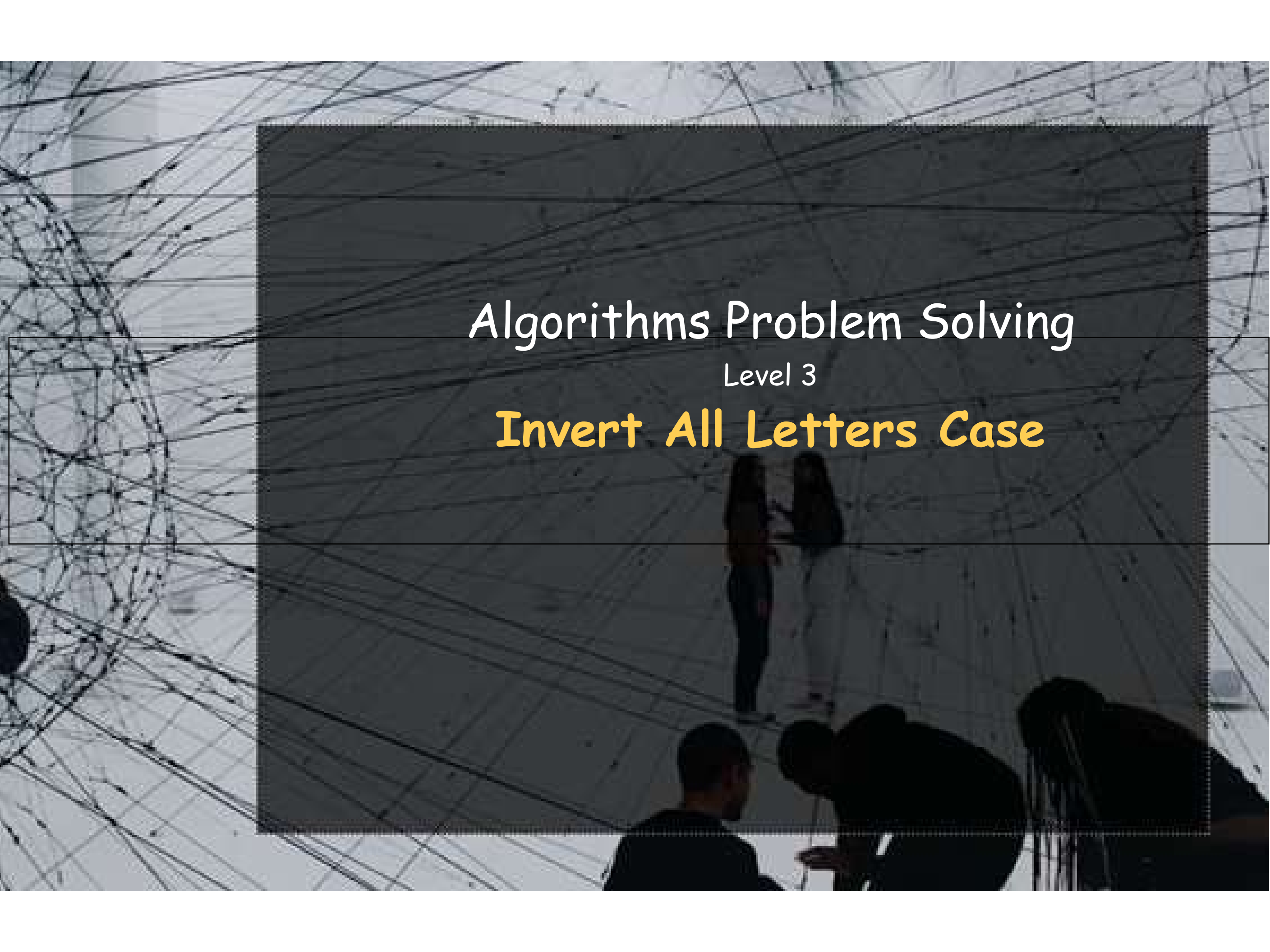


Algorithms Problem Solving

Level 3

Invert All Letters Case



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Definition

Full String Case Inversion

-  Program reverses the case of every alphabetic character in a string.

-  It reuses `InvertLetterCase()` to perform the transformation character by character.

Explanation

Complete String Traversal

-  Every character in the input string is visited.

-  No character position is skipped.

Helper Function Reuse

-  Each character is passed to ``InvertLetterCase()``.

-  The returned character replaces the original character at that position.

Structural Preservation

-  Spaces, digits, and punctuation remain in their original positions.

-  Only the alphabetic case is changed.

Procedures

1. Read the complete input string.
2. Traverse every character.
3. Call `InvertLetterCase()` for each character.
4. Store the returned value.
5. Print the inverted string.

💡 Example

Input:

Sara Alterkawi

➡ Result

sARA aLTERKAWI

Note

- ◆ The function demonstrates code reuse because it builds directly on the helper from Problem #27.
- ◆ The operation is reversible: applying the same transformation twice restores the original case.
- ◆ String length remains unchanged.

⊘ Warning

- ◆ The helper function must return the converted character correctly.
- ◆ The loop must remain within valid string indices.
- ◆ Basic ``<cctype>`` functions do not provide complete Unicode case handling for all languages.



Comparison

▣ `InvertAllStringLettersCase()`

- ✓ Toggles each letter individually.
- ✓ Preserves the original positions of spaces and symbols.
- ✓ Uses the single-character helper function.

▤ `UpperAllString()` / `LowerAllString()`

- ✓ Force all applicable letters to one specific case.
- ✓ Do not preserve the original upper/lower distinction.
- ✓ Are not reversible without storing the original string.

✦ Summary

- ✓ Traverses the entire string.
- ✓ Reuses `InvertLetterCase()` for every character.
- ✓ Changes lowercase letters to uppercase and vice versa.
- ✓ Preserves string structure and non-alphabetic characters.

Thank you 😊